

Content:

1. **Understanding the state of the art**
 2. **Sources of information**
 3. **Evaluation of sources**
 4. **Accessing information**
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1. Understanding the state of the art

For the researcher, it is important to establish an understanding of *state-of-the-art*. A researcher must have good understanding of the field or area of research to avoid falling onto the following problems:

- The intended study has been done numerous times before and there are masses of publications
- There are several recent developments that were not identified leading to flawed or outdated reporting
- Studies exist that collectively have successfully solved the problem
- The sources of information are not recent and hence, no longer relevant
- Reinventing the wheel, instead of using recent publications as a *springboard*

A quality literature study is key to avoiding the above shortfalls. Goddard and Melville (2001) define two types of literature review namely, preliminary literature review and full literature review. The **preliminary literature** review allows the researcher to *get a feel* for the field of enquiry. It is usually done as part of an application for Masters or PhD usually in the form of a concept note. The reading might be from more accessible sources. The purpose is to identify the main researchers or contributors to the field, to determine the most recent developments, and establish what further sources of information are needed in the full literature review. The bulk of the **full literature review** must be done before empirical work begins. The purpose is to provide motivation for the need for the research via critical analysis of the *state-of-the-art*.

Information overload! There are many publications contributing to a given field. The researcher has to identify the key authors, seminal works and select only those sources.

2. Sources of information

There are various sources of information available to the researcher. Some of these include:

- Journal articles
- Magazine articles
- Newspaper articles
- Text books
- Reference tools like online dictionaries and encyclopaedias
- Technical reports
- Conference proceedings

- Conference posters
- Dissertations/thesis
- Websites
- Personal communication

3. Evaluation of sources

The researcher has many sources of information most of which are easily available. The most reliable sources of information are journal articles, conference proceedings and thesis and dissertations. The above sources of information have some degree of peer review.

Peer review is when:

“the editor of the journal will identify a few other scholars (peers) who have expertise on the subject and send a copy of the article to those individuals. The peers will review the article for quality and rigor and give the editor their comments and recommendations. Depending on what the peers say, the editor may reject the article (if the peers found it to be substandard), or ask the author to revise and improve the article based on recommendations the peers provided” (Moritz, Vassar College Library, 2019).

Conference proceedings undergo a similar process of review. The dissertation/thesis of Masters and PhD students have been examined by at least two experts in the field that have been appointed by the University.

Differentiate between popular sources and scholarly sources of information.

The characteristics of *popular sources* of information include:

- Written in a style intended for a general audience (school learner, student)
- The writing style is informal and easy to understand
- The author/s affiliations and credentials are not known. Is this really an expert?
- Little to no citations to primary sources of information
- No peer review
- Biases may be apparent

Scholarly sources have the characteristics of:

- Authors include academic affiliations and credentials
- Will conform to a general format (abstract—lit review—conclusion)
- Neutral and factual writing style
- Use of field specific vocabulary or nomenclature
- Clear indication of sources of information that support the work. Include in-text citations and a reference list.
- The intention is to advance the field by contributing new knowledge
- Peer reviewed
- Trustworthy publishers (IEEE, ACM, Springer)

Differentiate between primary sources and secondary sources.

Primary sources of information are those that proclaim new findings or knowledge. When researchers produce new knowledge this is shared with the research community. These are used to advance the field of enquiry (discourse) via scholarly sources. That is, the publication is peer reviewed to validate the findings and/or results. Primary sources can also include dissertations/thesis.

Secondary sources of information provide a simplification, summary or analysis of several primary sources. A *literature review* is a common secondary source of information. A literature review is a synthesis of literature to provide a view on the state of the field.

Accredited list of publications *versus* predatory publications.

Accredited publications are those publishers that are recognized by international quality standards bodies such as, Web Of Science, Scopus and Elsevier. RSA has the DHET accredited list of publishers updated each year. Accredited publishers have high quality standards and expert reviewers. They are trustworthy sources of information and the University and researcher gets awards for publishing with them

Predatory publishers are nothing more than a money making scheme. Publications are of very poor quality and have no reputation among academics and the scientific community. Often resort to devious methods for trapping researchers and authors. Use Cabells listing available from the DUT library website to see listings of predatory journals.

4. Accessing information

The researcher needs to adopt a methodical approach to accessing information and gauging its quality. One approach is to:

1. Find sources (preferably journals, texts and conference proceedings)
2. Determine the quality of the content
3. Utilise a tool to organize the sources according to different sub-sections of the study

The relevancy of the source can be gauged by:

1. title
2. abstract
3. number of citations
4. impact factor of the journal

DUT library has subscriptions to many online journals, masters and PhD dissertations and online texts.

There will be a scheduled demonstration by DUT library during a lecture.

5. Software tools

Managing the references in a research work can become an overwhelming task. If mistakes are made the integrity of the write-up is compromised and in the worst case, the researcher can be accused of plagiarism. A manual process for this task is dangerous especially for thesis and dissertations.

Citation management tools assist the researcher to create local libraries of sources of information that they can reference, sort, share and organize their research work. Typically, this software bolts-on to a word processing package.

There are several citation management tools to assist the researcher to manage their sources of information. These are a must have for the researcher. RefWorks, EndNote, Zotero, Mendeley, and F1000 Workspace are popular tools.

These tools are used to assist the researcher to manage their sources of information and maintain a high standard of referencing in their work. The major uses include:

- organizing references
- citations can be directly imported from databases or search engines
- search databases for articles on a particular topic
- generate bibliographies/reference lists
- change the reference system (IEEE, Harvard)
- file management (e.g. pdf articles)
- networking with peers to share articles, libraries of references

Tutorials

1. Differentiate between scholarly and popular sources of information.
2. What is predatory publication? Go to the Cabells listing on the DUT website and identify three predatory IT publishers.
3. List three accredited IT journals. See the DHET list.
4. Provide 2 points of motivation for the importance of peer review process.
5. Briefly describe the peer review process.
6. Provide 3 reasons that a post-graduate student should use a citation management tool.
7. Is a PhD thesis an example of a primary source of information? Explain.

References

Goddard, W. and Melville, S. 2001. Research methodology: an introduction. Second edition. Juta.

Ivey, C., & Crum, J. (2018). Choosing the Right Citation Management Tool: Endnote, Mendeley, Refworks, or Zotero. Journal of the Medical Library Association: JMLA, 106(3), 399–403. doi:10.5195/jmla.2018.468.

<https://library.vassar.edu/c.php?g=510337&p=3489719>

<https://www.tutorspoint.com/blog/revealing-5-important-benefits-of-using-citation-tools/>